 Mensuration

Mensuration

Covering

- Units of measure
- Area and perimeter
- Circles, arcs and sectors
- Surface area and volume
- Compound shapes and parts of shapes

Units of measure

Length

- 1 km = 1000 m
- 1 m = 100 cm
- 1 cm = 10 mm

- 1 km^2 = 1000000 m^2
- 1 m^2 = 10000 cm^2
- 1 cm^2 = 100 mm^2

Volume

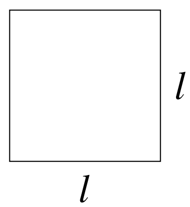
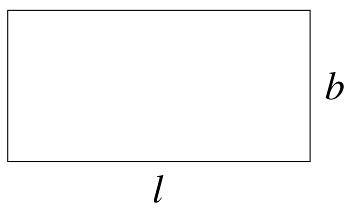
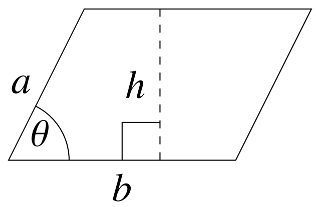
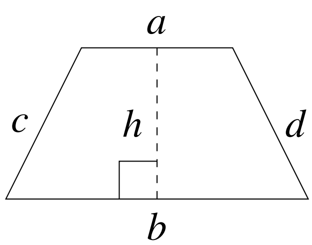
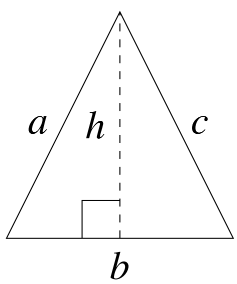
- 1 ℓ = 1000 cm^3
- 1 m^3 = 1000 ℓ
- 1 kℓ = 1000 ℓ
- 1 ℓ = 1000 mℓ

Mass

- 1 tone = 1000 kg
- 1 kg = 1000 g

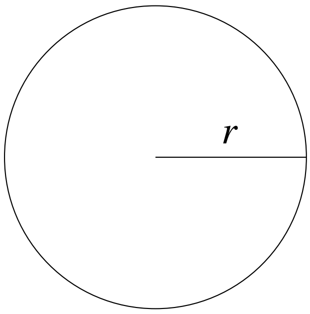
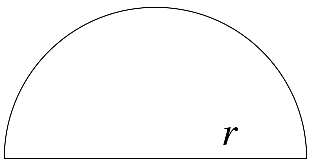
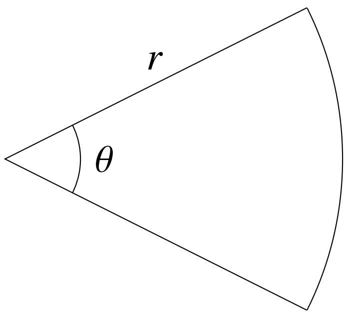
 Mensuration

Area and Perimeter

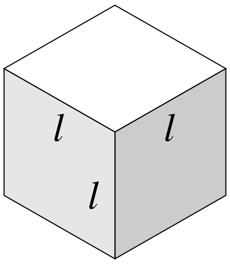
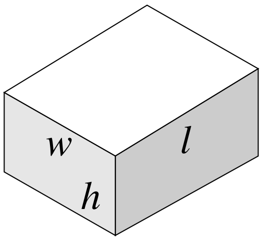
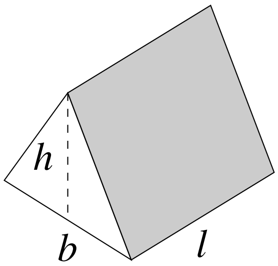
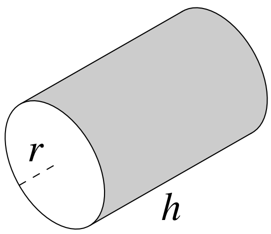
Name	Diagram	Area	Perimeter
Square		Area $= l \times l$	Perimeter $= 4 \times l$
Rectangle		Area $= l \times b$	Perimeter $= 2(l + b)$
Parallelogram		Area $= b \times h$ or Area $= ab \sin\theta$, Where a, b sides θ is angle by sides.	Perimeter $= 2(a + b)$
Trapezium		Area $= \frac{1}{2}(a + b)h$	Perimeter $= a + b + c + d$
Triangle		Area $= \frac{1}{2} \times b \times h$ or Area $= \frac{1}{2}ab \sin C$ For C is angle by sides a and b.	Perimeter $= a + b + c$

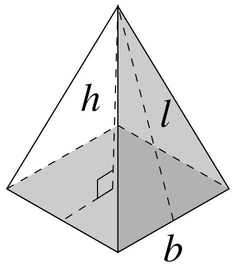
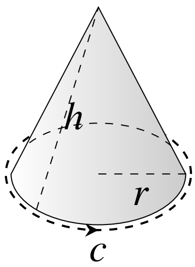
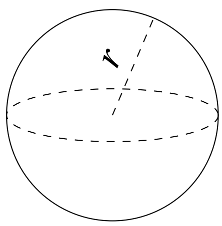
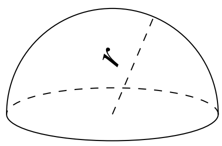
 Mensuration

Circles, arcs and sectors

Name	Diagram	Area	Perimeter
Circle		Area $= \pi r^2$	Circumference $= 2\pi r$
Semicircle		Area $= \frac{1}{2}\pi r^2$	Perimeter $= \pi r + 2r$
Sector		Area $= \dfrac{\theta}{360} \times \pi r^2$	Length of Arc $= \dfrac{\theta}{360} \times 2\pi r$ Sector Perimeter $=$ Arc $+ 2r$

Surface area and volume

Name	Diagram	Volume	Surface Area
Cube		Volume = l^3	Surface Area $= 6l^2$
Cuboid		Volume = $w \times h \times l$	Surface Area $= 2(lw+lh+wh)$
Prism		Volume = base area × height For triangle prism $= \frac{1}{2}bh \times l$	Surface Area = 2 × base area + (perimeter × height) For triangle prism $= bh + l(a+b+c)$
Cylinder		Volume = $\pi r^2 h$	Surface Area $= 2\pi r(h+r)$ Curved Surface Area $= 2\pi rh$

Name	Diagram	Volume	Surface Area
Pyramid		Volume $= \frac{1}{3} \times$ base area $\times$ height For square pyramid $= \frac{1}{3}b^2h$	Surface Area $=$ base area $+ \left(\frac{1}{2} \times l \times \text{base perimeter}\right)$ For square pyramid $= b^2 + 2bl$
Cone		Volume $= \frac{1}{3}\pi r^2 h$	Surface Area $= \pi r(r + l)$ Curved Surface Area $= \pi r l$
Sphere		Volume $= \frac{4}{3}\pi r^3$	Surface Area $= 4\pi r^2$
Hemisphere		Volume $= \frac{2}{3}\pi r^3$	Surface Area $= 3\pi r^2$ Curved Area $= 2\pi r^2$

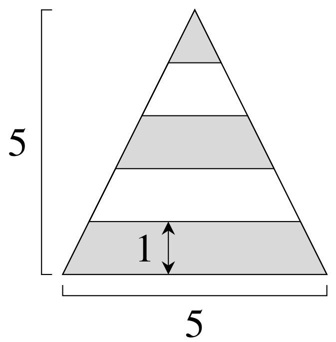
Compound shapes and parts of shapes

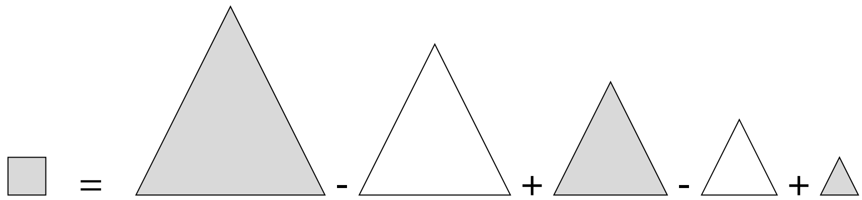
Compound shapes for 2d:

To find wanted area we add or subtract areas.

To find perimeter we find and add up all the outer lengths.

E.g





$$A = \frac{1}{2}(5^2 - 4^2 + 3^2 - 2^2 + 1^2) = 7.5$$

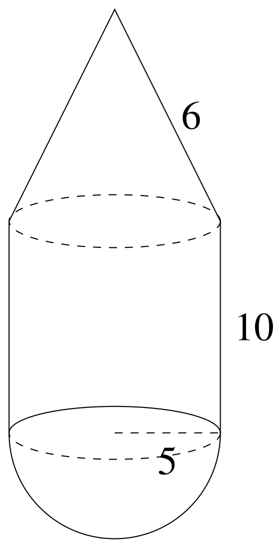
Compound shapes for 3d:

To find wanted volumes, we add or subtract other volumes.

To find surface areas, we have to be cautious of which faces are covered or not.

E.g, The covered faces are all the circles, hence we won't count them in the final formula.

$Surface\ Area = \pi r l + 2\pi r h + 2\pi r^2 = \pi(5 \times 6 + 2 \times 5 \times 10 + 2 \times 5^2) \approx 565.487$

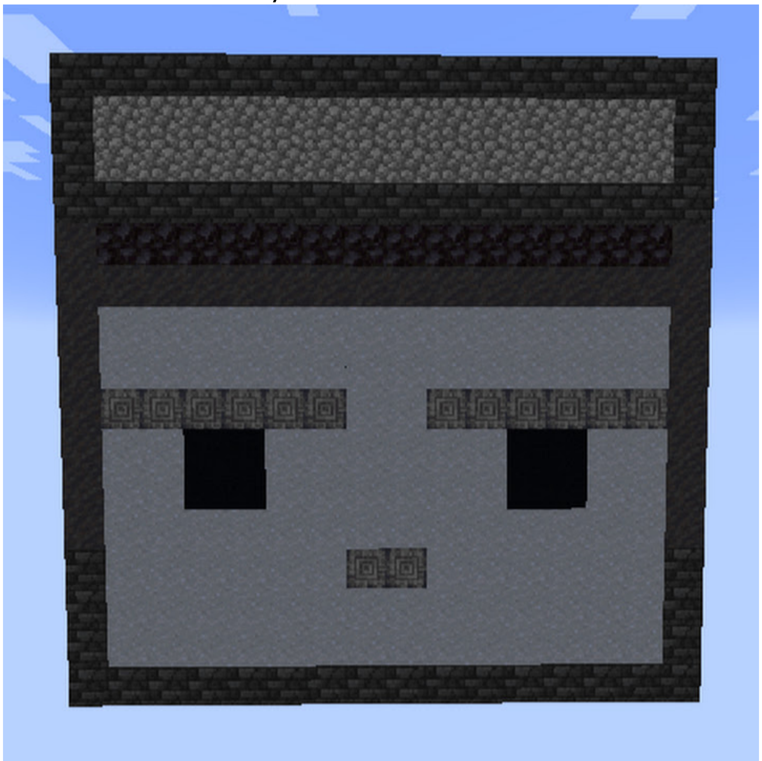


Mathematics is a beautiful subject—logical, creative, and deeply rewarding. It is truly worth investing your time in, not just for exams, but for the way it trains your mind to think clearly and solve problems with confidence. To excel in mathematics, you don't need a natural passion for the subject; what matters most is motivation and consistency. Progress comes from practice, patience, and the willingness to learn from mistakes. There will be challenges, but hard work, persistence, and never losing hope make all the difference. Keep trying, even when it feels difficult—every attempt strengthens your understanding. These notes are meant to reinforce that mindset and support your journey through **Mathematics O Levels and IGCSE**, reminding you that improvement is always possible with effort and determination.

If you don't know the method to study efficiently, try looking at the 67 Day Plan in the Study Material folder

Here

Checkout my channel @HarisRehmanGG



Where to find me:
Discord `haris_rehman`